Group Chartreuse

Agenda: We want to discuss our modeling approach and have a few confusions regarding target definition so we would like to spend time discussing what definition is the more correct way of modeling.

Our modeling approach:

1. Create proper target definitions
2. Feature engineering
3. Feature selection
4. Modeling
5. Evaluation

Target definition:

Currently we have 2 target definitions in mind:

1. We take default within 1 year of the statement date
2. We take eventual default (default can happen after 1 year of statement as well)

Target definition	Pros	Cons
1	1. Matches with the exact business requirement of providing 1 year PD	1. Very few defaults within a year (2.7% in the sample data) 2. Doing a univariate feature selection (IV) step won't be straight forward for the entire dataset. We will have to do it fs_year wise
2	1. Defaults are more (9.3% in sample data) 2. Feature selection is straightforward	1. Need calibration for 1 year PD

- In case of eventual default a firm will have multiple entries (rows) with target 1. This will be helpful for model to understand the sequential nature from the raw data itself
- In case of default within 1 year we will have to think of what to do with the other entries, for example: a firm defaulting in 2008 will have a datapoint of 2007 (statement year) with 1 as target but 0 for previous years. So if we define the target in this way then the model will see both targets 0 and 1 for the same firm which can create misunderstanding while training.

Feature engineering:

We believe the most important part is feature engineering. We would focus more on feature engineering rather than modeling. Following are the list of features ideas that we thought of implementing.

Link to features sheet: Features list

Feature selection:

We have thought of a fixed approach so far for feature selection in the following order:

1. Descriptive statistics (univariate)
2. IV (univariate)
3. CSI (drift analysis) (univariate)
4. Correlation (Bivariate)
5. RFE (multivariate)

We planned this order as performing univariate analysis will be fasters and we can drop features with less predictive power before going onto compute expensive bivariate / multivariate steps

Evaluation:

We have finalized the following metrics to evaluate models:

1. KS statistic (main class separation metric)
2. BCR (top 1 or 2 percentile) (capture rate)
3. AUC (check metric) must pass if KS is passing
4. Model PSI (must be less than a threshold, 20%)
5. Top feature importance (top 2-3 features combined importance must be less than <= X%)

Modeling:

We believe most of the importance will come from feature engineering rather than modeling. For modeling we would like to keep the models simple for explainability reasons. We hence decided to go for the Logit model as our baseline and build tree based models on top of it.

This is our current finalized framework and would like to get any inputs from your end.